

Program : Diploma in Instrumentation Engineering	
Course Code : 5089C	Course Title: Robotics and Automation Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week : 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To acquire the knowledge on algebraic tools for the description of motion.
- To develop the ability to analyze and design the motion for articulated systems.
- To develop an ability to use software tools for analysis and design of robotic systems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics II	2
Basic Knowledge of IT and basic problem solving		Introduction to IT systems	1
Sensors, Actuators		Sensors and Transducers	3
Different types of processes		Process Control Instrumentation	4

Course Outcomes:

On completion of the course, the student will be able to

COn	Description	Duration (Hours)	Cognitive level
CO1	Build basic MATLAB programs	14	Applying
CO2	Experiment with Cartesian coordinate and joint co ordinates.	9	Applying
CO3	Develop pick and place programs.	10	Applying
CO4	Build programs to operate and control a robot.	10	Applying
	Lab Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2			3				
CO3			3				
CO4			3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Build basic MATLAB programs		
M1.01	Simulate controlled, un controlled half wave and Full wave rectifiers	5	Applying
M1.02	Plot the Time response for standard test signals for first order system	4	Applying
M1.03	Build MATLAB program for Root locus and Bode plot	5	Applying
CO2	Experiment with Cartesian coordinate and joint co ordinates.		
M2.01	Robot operation, moving the various axis continuous and intermittent motions.	5	Applying
M2.02	Homing operation, Recording positions	4	Applying
	Lab Test - I	1	
CO3	Develop pick and place programs.		
M3.01	Writing programs for pick and place objects.	5	Applying
M3.02	Write Programs for stacking objects.	5	Applying
CO4	Build programs to operate and control a robot.		
M4.01	Write programs to transport items.	5	Applying
M4.02	Write programs to sort items according to colour, weight and height	5	Applying
	Lab Test - II	1	

Text / Reference:

T/R	Book Title / Author
R1	Mikell P. Weiss G.M., Nagel R.N., Odraj N.G., Industrial Robotics, McGraw Hill Singapore, 1996
R1	Ghosh, Control in Robotics and Automation: Sensor Based Integration, Allied Publishers, Chennai, 1998.
R2	Deb. S. R., Robotics technology and flexible Automation, John Wiley, USA 1992.
R3	Issac Asimov I Robot, Ballantine Books, New York, 1986.
R4	Thomas R. Kurfess Ph.D., P.E. Robotics And Automation Handbook CRC press

Online Resources:

Sl. No	Website Link
1	https://nptel.ac.in/courses/112/101/112101098/
2	http://www.nptelvideos.in/2012/12/robotics.html
3	https://www.lccc.edu/academics/science-and-engineering/science-in-motion/labs-equipment/robotics-lab-experiments

